

AEGIS: Matrix-Free Operator — Audit, Certify, and Defend Graph Neural Networks

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Abstract

Deploying a graph neural network safely takes two things that current tools provide only in isolation: a map of which edge perturbations would break its predictions, and a robustness guarantee a practitioner can act on. AEGIS delivers both from a single matrix-free object, the *constrained sensitivity matrix* S_c , which records how strongly each edge can move the model’s equilibrium prediction and scales to $N=7,650$ nodes on one GPU. From this one object come three capabilities. AEGIS *audits*: its leading singular direction, column norms, and node margins give an optimal attack, an edge-vulnerability ranking, and per-node safe radii in one pass. It *certifies*: AEGIS-Conformal bounds in closed form how far a bounded perturbation can shift the conformity scores, turning split conformal prediction into a distribution-free guarantee that holds at the nominal level under the very attack it certifies, with no Monte-Carlo sampling. This certificate stays non-vacuous where randomized smoothing degenerates to the full label set at $10^4\times$ the cost, and for contractive models a closed-form safe radius adds a deterministic guarantee that the empirical breaking budget exceeds by $2\text{--}9\times$. It *defends*: penalizing the leading sensitivity trades clean accuracy for certified robustness. Empirically, this continuous sensitivity predicts discrete attack damage across 6 datasets, 7 architectures, and 4 domains (rank correlation $\tau=0.99$), and one analytic query inflicts $74\text{--}156\times$ the per-query damage of a 50-step PGD attack.

1 Introduction

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges (Zügner, Akbarnejad, and Günnemann 2018), drug-interaction models misfire under perturbed molecular graphs (Dai et al. 2018), and power grids can miss critical contingencies when their graph representations are structurally fragile (Nakiganda et al. 2023). A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*. Structural attacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021; Wu et al. 2019) rank edges by surrogate gradients, but return no continuous direction and no per-node

budget. Smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023, 2021) and deterministic certifiers (Li and Wang 2025; Zügner and Günnemann 2019b) return per-node certificates, but neither an edge ranking nor a direction. Robust-architecture defenses (Zhu et al. 2019; Zhang and Zitnik 2020; Jin et al. 2020) harden models without ever surfacing which edges are vulnerable, and sober-look critiques (Mujkanovic et al. 2022; Gosch et al. 2023) catalogue the evaluation pitfalls that follow. No prior method audits, certifies, and defends in a single matrix-free pass. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) does, and all three capabilities come from one analytical object: the constrained sensitivity matrix S_c (section 3). The idea is simple. S_c records how strongly each edge can push the model’s equilibrium prediction, so its leading singular direction, its column norms, and its per-node margins deliver the three diagnostics together, from one randomized SVD over the equilibrium resolvent rather than three separate analyses, and the same analytic bound upgrades each per-node radius into a distribution-free certificate (section 4.1). Two scope boundaries are worth stating up front. The closed-form characterisation of when a model breaks (Theorem 1) holds for contractive implicit GNNs, whereas the operator S_c itself extends to any GNN with continuous edge-weight-modulated message passing (section 5.1); throughout, the threat model is edge deletion and weight perturbation (section 2).

Contributions. (1) **Constrained sensitivity operator.** S_c specialises equilibrium IFT sensitivity (Koh and Liang 2017; Gould, Hartley, and Campbell 2021) to *structural* edge perturbations via the projection P_c . One matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three; it matches a dense σ_1 to 0.03% at $N=200$ and scales to $N=7,650$. (2) **Certification from sensitivity.** The same bound yields two guarantees. AEGIS-Conformal turns it into a *distribution-free* certificate: coverage $\geq 1-\alpha$ over the ε -ball, holding at the nominal level under worst-case attack and staying non-vacuous even where the deterministic radius is thin (section 4.1). For contractive models we add a constant-factor two-sided characterisation of the breaking point $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$, whose norm certificate empirically under-states the true break by $2\text{--}9\times$ (10 seeds, section D.4). (3) **Coupled defense.** Penalizing $\sigma_1(S_c)$ trades

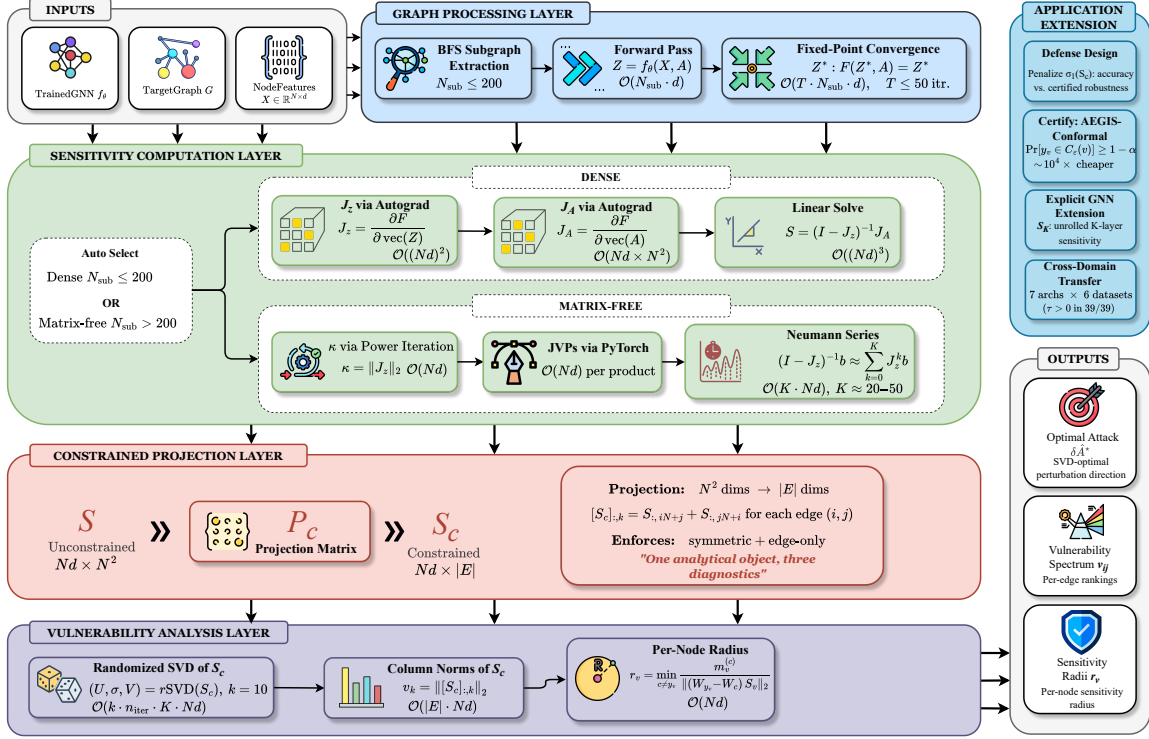


Figure 1: The four-stage AEGIS pipeline (section 3): (1) graph processing (BFS subgraph or full-graph fixed-point pass); (2) sensitivity (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) analysis (randomised SVD, S_c column norms, per-node radii).

clean accuracy for certified robustness (larger certified radii, higher Conformal coverage), so the operator that finds the worst attack also tunes the defense (section 5.3). **(4) Empirical evaluation.** 6 datasets, 7 architectures, 4 domains (390 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD (Geisler et al. 2021) with complementary positioning against AGNNCert (Li and Wang 2025) (section D.2), continuous-to-discrete transfer (fig. 5), and a fraud-detector audit (section 6).

2 Preliminaries and Threat Model

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes, adjacency $A \in \mathbb{R}^{N \times N}$, features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$. Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, $\text{vec}(\cdot)$ column-major vectorization. The sensitivity matrices (S, S_c, S_K, S_v) and governing scalars ($\kappa = \|J_z\|_2, \rho(J_z), \eta, \varepsilon_{\text{crit}}, r_v$) are defined where they first appear.

IGNN and IFT. AEGIS analyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN (Bai, Kolter, and Koltun 2019; Bai, Koltun, and Kolter 2020) iterates a weight-tied F_θ to a fixed point; the IGNN instantiation (Gu et al. 2020) uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized (Miyato et al. 2018)

so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation through the fixed point (Gould, Hartley, and Campbell 2021; Fung et al. 2022) with optional Jacobian regularization (Bai, Koltun, and Kolter 2021), and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}. \quad (2)$$

Equation (2) is the lever AEGIS pulls: it gives the closed-form first-order change in z^* under any parameter p . Here the resolvent $(I - J_z)^{-1}$ acts as a gain that amplifies the raw perturbation by at most $1/(1 - \kappa)$ through its convergent Neumann series, so the closer the model drifts to losing contractivity ($\kappa \rightarrow 1$), the more violently its equilibrium can move. The phase transition of section 4 makes this intuition precise. One refinement matters there: the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ (Trefethen and Embree 2005) measures how far the true gain sits above its spectral-radius estimate, equalling 1 exactly when J_z is normal.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta \hat{A}, \quad \|\delta \hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta \hat{A}$ symmetric ($\delta \hat{A} = \delta \hat{A}^\top$), preserving the undirected

graph), edge-only ($[\delta\hat{A}]_{ij}=0$ for $(i,j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge deletion and weight perturbation. Insertion attacks (Nettack-class (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a) and node injection (Sun et al. 2020)) would require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that leaves the S_c construction intact. Discrete deletions connect to the continuous model through Proposition 3. Unlike input-feature perturbation (El Ghaoui et al. 2021; Revay, Wang, and Manchester 2020), structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F / \partial \text{vec}(A)$, motivating the S_c construction of section 3.

3 The AEGIS Operator: One Object, Three Diagnostics

The previous section isolated the two Jacobians, J_z and J_A , through which the adjacency moves the equilibrium. AEGIS packages them into a single operator, the constrained sensitivity matrix S_c , and reads every diagnostic off it. One S_c computation returns the per-edge vulnerability spectrum $\{v_{ij}\}$, the SVD-optimal attack direction $\delta\hat{A}^*$, and the per-node safe radii $\{r_v\}$, together with $\varepsilon_{\text{crit}}$ and the regime characterisation of Theorem 1 for IGNN-class operators. None of this requires building S_c explicitly: the pipeline only ever applies it to vectors, so it never materialises $S \in \mathbb{R}^{Nd \times N^2}$ or forms the resolvent. A dense path ($N \leq 200$) takes an exact SVD; a matrix-free path ($N > 200$, validated to $N=7,650$) uses JVP/VJP queries and a randomized SVD. Figure 1 illustrates the four stages and algorithm 1 the routing between the two paths.

Algorithm 1: AEGIS matrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Require: F , fixed point z^* , $\hat{A}$, edge set E , budget ε , rank k , Neumann tolerance tol ; (opt.) head W , margins $m_v^{(c)}$.

Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^*$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.

- 1: $\kappa \leftarrow \|J_z\|_2$; $K \leftarrow \lceil \log(1/\text{tol}) / \log(1/\kappa) \rceil$
- 2: $S_c v := \left(\sum_{j=0}^K J_z^j \right) J_A P_c v$ {matrix-free}
- 3: $(\sigma_1, v_1) \leftarrow \text{rSVD}(S_c, k)$ (Halko, Martinsson, and Tropp 2011)
- 4: $\delta\hat{A}^* \leftarrow \varepsilon \text{sym}(P_c v_1) / \|\text{sym}(P_c v_1)\|_2$
- 5: $v_{ij} \leftarrow \|S_c e_{ij}\|_2$ for each $(i, j) \in E$
- 6: $r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c) S_{c,v}\|_2$ {head only}
- 7: $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$ {Theorem 1}
- 8: **return** $(\sigma_1, \delta\hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$

The constrained sensitivity operator. Intuitively, S_c is the full sensitivity S cut down to the perturbations the threat model actually permits: one column per existing edge, symmetrised so that touching edge (i, j) also touches (j, i) . Formally, $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ in the edge basis $b_k = (e_i e_j^\top + e_j e_i^\top) / \sqrt{2}$; each b_k has unit Frobenius norm, so $\sigma_1(S_c)$ is measured consis-

tently with the budget $\|\delta\hat{A}\|_F = \varepsilon$. The $N^2 \rightarrow |E|$ projection P_c that performs this restriction is the standard duplication-matrix reduction (Magnus and Neudecker 2019) onto the edge-supported symmetric subspace. A single randomized SVD of S_c then reads off all three diagnostics: (i) the optimal attack direction (Proposition 1); (ii) the per-edge vulnerability ranking $v_{ij} = \|[S_c]_{:,k}\|_2$; and (iii) the per-node safe radius r_v , whose first-order guarantee we prove in Proposition 2 (section 4). Together these three readings are what it means to *audit* the model. The same operator then *certifies* it, because the radius upgrades to a distribution-free guarantee (section 4.1), and *defends* it, because penalizing $\sigma_1(S_c)$ trades clean accuracy for certified robustness (section 5.3).

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S , with shift $\varepsilon \cdot \sigma_1(S)$. Restricted to the threat model’s feasible set (symmetric, edge-supported; section 2) the maximizer is instead the leading right singular vector of the constrained S_c in the edge basis $\{b_k\}$, with shift $\varepsilon \sigma_1(S_c) \leq \varepsilon \sigma_1(S)$; the unconstrained $\text{reshape}(v_1)$ is generally infeasible.*

This direction maximises hidden-state shift rather than label change, so it pinpoints the most dangerous way to spend the budget rather than certifying a global safety margin (section 5.1); the gap to that margin is the pseudospectral envelope of Observation 1 and Remark 1. Empirically the direction nonetheless aligns with classification damage (section 5.1).

Matrix-free operator. Scale rests on never building S_c : we only ever apply it to vectors, which costs $O(Nd)$ memory instead of the $O((Nd)^2)$ a dense S_c would need. The key product $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated with a truncated Neumann series $\sum_{k=0}^K J_z^k b$, whose depth K adapts to κ and stops early once $\|J_z^k b\| < 10^{-6} \|b\|$. Every Neumann term, like the product $J_A \text{vec}(\delta A)$, is a forward-mode JVP costing $O(Nd)$ with P_c applied implicitly; the dense path ($N \leq 200$) instead materialises J_z, J_A and solves directly. The randomized SVD (Halko, Martinsson, and Tropp 2011) ($k=p=10$, $n_{\text{iter}}=2$) returns (u_1, σ_1, v_1) , and symmetrising through $\text{sym}(M) = (M + M^\top)/2$ keeps the recovered attack $\delta\hat{A}^* = \varepsilon \text{sym}(P_c v_1)$ inside the threat model (section 2). One query suffices because the singular spectrum is well separated (gap $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39 - 0.50$; fig. 2), and that separation is what lets a single rSVD match iterative attackers (section 5.1).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section D.4).

4 Adversarial Equilibrium Theory

Section 3 read three diagnostics off the operator S_c ; we now establish the guarantees that justify them, beginning with the safety boundary. The governing picture is a phase transition in the perturbation budget ε . While ε stays small, the perturbed model remains contractive and its prediction moves

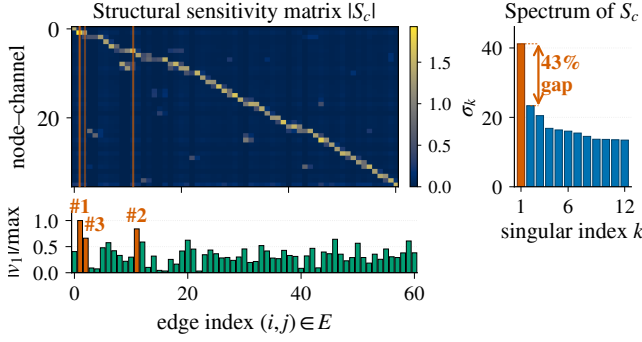


Figure 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50$, $d=16$, $|E|=61$, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum ($\sigma_1=41.2$, 43% gap to σ_2).

only boundedly with the graph. As ε climbs toward a critical budget $\varepsilon_{\text{crit}}$, the equilibrium grows arbitrarily sensitive, and once ε passes $\varepsilon_{\text{crit}}$ contraction is no longer guaranteed and the prediction can break outright. Theorem 1 makes this picture precise and puts $\varepsilon_{\text{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT (Bolte and Pauwels 2021) (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14\text{--}0.59$ across our suite; table 1). (A2) does not imply (A3); we report the trained κ so practitioners can audit, and AEGIS still returns rankings and attacks if (A3) fails, its guarantees then scoped to the contractive regime.

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

Then structural perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ exhibit three regimes:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \quad (5)$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}}/(1 - \kappa)$.

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): The resolvent obeys $\|(I - J'_z)^{-1}\|_2 \geq 1/\min_i |1 - \lambda_i(J'_z)|$, blowing up when an eigenvalue of J'_z approaches 1 (a vanishing spectral gap to 1, not $\|J'_z\|_2 \rightarrow 1$ alone). The rate is $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ when J'_z is normal with a dominant real-positive (Perron) eigenvalue; in general $\varepsilon_{\text{crit}}$ lower-bounds the divergence threshold, the slack being the nonnormality η (Observation 1; $\eta \leq 2.47$ for general ReLU, Remark 1).

(c) **Supercritical** ($\varepsilon \geq \varepsilon_{\text{crit}}$): The contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1); at the worst-case

$\kappa = \|\hat{A}\|_2 \|W\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ is a sufficient safety boundary that trained IGNNs sit well inside (a 2–4 $\times$ spectral-radius margin to $\rho=1$, section D.4).

Proof in section A.1. The subcritical bound is a conservative IFT (Bolte and Pauwels 2021) with Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$; the critical rate follows from the resolvent identity, which $\varepsilon_{\text{crit}}$ lower-bounds (slack η).

The budget $\varepsilon_{\text{crit}}$ is only a closed-form *sufficient* bound, one that the data-dependent S_c tightens by 7–14 $\times$. The slack between them is precisely the Neumann-versus-spectral-radius gap, the pseudospectral index η , which for symmetric $\hat{A}$ is controlled by W alone ($\eta \leq \kappa(V_W)$ in the all-active case, $\eta \in [1.19, 2.47]$ for ReLU; Observation 1 and Remark 1).

Theorem 1(a) certifies safety *below* $\varepsilon_{\text{crit}}$; the same operator bounds the boundary from *above*, through the nonnormality $g_W = \|W\|_2 / \rho(W) \geq 1$ of W (the *entire* norm-vs-radius gap for symmetric $\hat{A}$).

Theorem 2 (Constant-factor two-sided boundary, all-active). *Under (A1)–(A3) with $\hat{A}$ symmetric and $\varepsilon_{\text{crit}} > 0$, the all-active contraction boundary ε_{br} (the smallest feasible $\|\delta\hat{A}\|_F$ at which $\rho(\hat{A}' \otimes W) \geq 1$) obeys*

$$\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}} \leq \frac{C}{\beta} \varepsilon_{\text{crit}}, \quad C = g_W \frac{1 + \kappa}{1 - \kappa}, \quad (6)$$

with C dimension-free and computable post-training, and $\beta \in (0, 1]$ the edge-supported Frobenius mass of $\hat{A}$'s Perron mode (positive for any connected graph). The upper side is attained by a rank-one critical-driving attack aligned with that mode; the bracket collapses to equality iff $g_W=1$ and $\beta=1$.

The constant is admittedly loose, with proven slack of $\sim 10\text{--}16\times$ on our suite, but its worth is the coupling it exposes: a single operator S_c supplies the certified budget $\varepsilon_{\text{crit}}$, the matching attack that approaches it, and, through $\sigma_1(S_c)$, the defense. Empirically the norm certificate under-states the true breaking budget by 2–9 $\times$ (10 seeds); the proof and a nonlinear extension are in section B.

The third diagnostic, the per-node radius r_v named in section 3, carries a first-order safety guarantee.

Proposition 2 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c) S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta\hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v ; substituting $S_{c,v}$ gives the tighter constrained radius $r_v^{(c)} \geq r_v$. Minimizing over all competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes.

Proof in section A.3. A Cauchy–Schwarz bound on each per-competitor margin change gives r_v as the smallest margin

divided by two amplification factors (decision-boundary and equilibrium sensitivity); high-margin, low-sensitivity nodes get large radii.

4.1 AEGIS-Conformal: A Distribution-Free Certificate

The radius r_v of eq. (7) is a first-order threshold: locally tight, but liable to be violated at larger ε . We now upgrade it to a distribution-free guarantee, one that is sound under an exchangeability condition and gated by an explicit coverage test (section C). The starting point is split conformal prediction, which returns a set $C(v)$ with $\Pr[y_v \in C(v)] \geq 1-\alpha$ on clean inputs. Making this hold *robustly* over the ball $\|\delta\hat{A}\|_F \leq \varepsilon$ requires bounding the worst-case shift in the conformity scores, and that is exactly what the same S_c sensitivity delivers in closed form: a curvature-corrected bound $L_1^c\varepsilon + C_v\varepsilon^2$ that replaces the $\sim 10^4$ -sample smoothing the task would otherwise demand. Substituting it into the binary-certificate split conformal of Zargarbashi, Antonelli, and Bojchevski (2023) yields a robust set with

$$\Pr[y_v \in C_\varepsilon(v)] \geq 1 - \alpha \quad \text{for all } \|\delta\hat{A}\|_F \leq \varepsilon. \quad (8)$$

Section 5.2 validates this certificate empirically, checking both nominal-level coverage and the coverage gate under the worst-case attack, and section 5.3 returns to the coupled $\sigma_1(S_c)$ defense.

4.2 Transfer to Discrete Edits and Explicit GNNs

Two bridges connect this continuous operator to the discrete, architecture-general setting practitioners actually face. The first carries the continuous score to discrete edge removal: modelling a single-edge deletion as fixed-normalization masking, the discrete damage equals the edge-weighted sensitivity score plus a bounded second-order remainder, so for subcritical edges the score $A_{ij}v_{ij}$ preserves the brute-force N-1 ranking (Proposition 3), reaching $\tau=+0.996$ on full-graph Amazon Photo (section 5.1). The second frees the construction from implicit models altogether: a K -layer explicit GNN gets the same S_c from an unrolled sensitivity S_K (Proposition 4), so 6 of our 7 architectures inherit the full pipeline, with only the closed-form $\varepsilon_{\text{crit}}$ restricted to the contractive subclass. Both propositions are stated and proved in section A.

5 Experiments

Our evaluation follows the three capabilities of section 3. AEGIS *audits* (section 5.1): one analytic query matches iterative attackers and its S_c ranking transfers to discrete edge removal across architectures. It *certifies* (section 5.2): AEGIS-Conformal coverage holds at the nominal level under the very attack it certifies, where randomized smoothing is vacuous. It *defends* (section 5.3): a $\sigma_1(S_c)$ penalty buys certified robustness. A deployed fraud detector (section 6) closes the loop on real data.

Setup. We evaluate AEGIS on 6 datasets across 4 domains with 10 seeds each (PyTorch (Paszke et al. 2019)); $\pm$ are seed standard deviations. IGNN (Gu et al.

Table 1: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ is the local critical budget. AtkAdv: AEGIS/random damage. Cov%: subgraph nodes certified first-order-safe at $\varepsilon=0.05$ ($r_v>0.05$).

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	83 $\pm$ 10
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	94 $\pm$ 5
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	76 $\pm$ 18
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	86 $\pm$ 8
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	78 $\pm$ 4

Table 2: One-query audit vs. iterative attackers: equilibrium damage at $\varepsilon=0.10$ (IGNN, 50-node subgraph, 10 seeds). Cls-PGD: classification-loss PGD; Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	AEGIS	Cls-PGD	Shift-PGD	Random
Cora	3.70 $\pm$.79	2.51 $\pm$.48	3.05 $\pm$.70	0.97 $\pm$.25
Citeseer	4.63 $\pm$.71	2.97 $\pm$.42	3.35 $\pm$.57	1.01 $\pm$.15
WikiCS	2.10 $\pm$.22	0.67 $\pm$.11	1.93 $\pm$.20	0.53 $\pm$.09

2020) uses $F(Z)=\text{ReLU}(\hat{A}ZW^\top + U(X))$, $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised (Miyato et al. 2018), Adam (lr 0.01). Datasets: Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), Amazon Fraud (Dou et al. 2020); baselines: smoothing (Gaussian $\sigma=0.01$), Mettack, GR-BCD/PR-BCD (Geisler et al. 2021). All seven architectures appear only in fig. 5 and table 8; the rest use IGNN. Full tables, ablations, and timing are in section D; reproducibility in section F.

5.1 Audit: One Query Finds the Worst Attack

Two prerequisites must hold: the trained models sit in the contractive regime, and AEGIS’s perturbations beat chance. Table 1 confirms both, verifying (A3) on every benchmark ($\kappa<1$) and reporting per-dataset $\varepsilon_{\text{crit}}$ alongside a $3.2\text{--}4.1\times$ attack advantage over random. Unless stated otherwise, IGNN experiments run on 50-node BFS ego-subgraphs centred on the highest-degree node and report *local* vulnerability, while graph-level assessment uses the matrix-free pipeline (section D.4); all bounds use $\kappa = \|J_z\|_2$.

One-query attack. AEGIS’s attack is analytic and costs a single query, not an optimisation loop: by construction the SVD direction maximises the first-order equilibrium shift (Proposition 1). Three findings show that this single query reflects the model rather than a gradient artifact. First, a transfer direction from a *separately trained* surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos=0.99$), against $44\pm 4\%$ for a 512-query black-box search. Second, per query the attack delivers **74–156** $\times$ the *equilibrium-shift* damage of a 50-step PGD (Madry et al. 2018) (table 2); classification-loss PGD inflicts 15–70% less at $50\times$ the cost, and GR-BCD pulls ahead only at the larger budgets it is designed for (section D.2). Third, per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$), and iterative recomputation closes at most 3 pp of the static-versus-greedy gap at $k\times$ the compute, so the static ranking stands as the headline.

AEGIS also beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv), and inflicts 3–10 $\times$ more equilibrium damage than Mettack (Zügner and Günnemann 2019a) in the early-warning regime (149/150 paired wins, $p<10^{-43}$). The discrete-removal curves against a label-aware Greedy attacker appear in fig. 4, with GR-BCD/PR-BCD detail in section D.2. As a check on the linearisation, the first-order shift bounds agree with the actual shifts to within 1% at small budgets, loosening to $\leq 39\%$ on citation graphs by $\varepsilon=0.20$.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screen (section 5.2). Figure 3 reports prediction-flip fractions vs. ε (95% CI bands): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% at $\varepsilon=0.20$; Pubmed is the right-skewed outlier (mean 10.3% at $\varepsilon=0.10$, 27.4% at $\varepsilon=0.20$).

Continuous-to-discrete transfer. The construction is not tied to implicit models. For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ by finite differences and build S_c exactly as before (Proposition 4); only the IGNN additionally carries the boundary of Theorem 1. The question that matters in practice is whether the continuous S_c ranking predicts the discrete, brute-force N-1 removal order. Ranking edges by the edge-weighted score $A_{ij}v_{ij}$ of Proposition 3, fig. 5 reports the Kendall τ over 6 datasets and 7 architectures (390 runs): every one of the **39/39** cells is positive, with median $\tau = +0.99$ ($p < 10^{-5}$), against 34/39 for the unweighted score. The agreement even survives leaving the contractive regime: on full-graph Amazon Photo ($N=7,650$, $\kappa \approx 1.00$) it reaches $\tau = +0.996 \pm 0.002$. Per-architecture detail, the phase-transition and scalability sweeps (the matrix-free path reaching $N=7,650$), and the ablations are in sections D.4 to D.6; the ablations show vulnerability to be *delocalized*, so S_c localises risk for triage rather than removing it.

5.2 Certify: Distribution-Free Coverage

The certificate of section 4.1 holds empirically. Table 3 reports coverage over 10 seeds and four datasets. The crucial quantity is the *gate*, the coverage retained under the worst-case AEGIS attack at magnitude ε : it sits at the nominal $1-\alpha=0.90$ at $\varepsilon=0.01$ on every dataset and turns conservative (0.94–1.00) at $\varepsilon=0.05$. The sets stay informative throughout, running ~ 1.0 – 3.5 labels and tightening exactly where the deterministic radius is thin, all at zero Monte-Carlo samples. Two conditions are stated plainly: exchangeability on a single transductive graph is an assumption for which the empirical gate is the evidence, and S_c is formed densely at $N=200$, leaving a matrix-free conformal pass to future work.

Table 4 pits AEGIS-Conformal against randomized smoothing: on the matched Frobenius ball smoothing is *vacuous* (full label set, zero certified nodes); only on a strictly larger per-coordinate ball does it certify, and then at **23,000–57,000 $\times$** the cost of the zero-sample S_c bound

Table 3: AEGIS-Conformal coverage (10 seeds; Cora, Citeseer, Pubmed, WikiCS; $\alpha=0.1$, target 0.90). Each ε block reports clean coverage / mean set size, then the bold **gate** (coverage under the worst-case attack at ε). Classes: 7/6/3/10 resp.; std over seeds ≤ 0.07 . Sets <1 reflect occasional empty (abstaining) sets under TPS.

Data	score	$\varepsilon=0.01$		$\varepsilon=0.05$	
Cora	APS	0.90 / 1.37	0.90	0.89 / 1.06	0.98
	TPS	0.88 / 0.95	0.90	0.89 / 0.99	0.98
Citeseer	APS	0.92 / 1.50	0.92	0.92 / 1.35	0.97
	TPS	0.91 / 1.20	0.92	0.92 / 1.26	0.96
Pubmed	APS	0.91 / 1.42	0.95	0.92 / 1.52	1.00
	TPS	0.90 / 1.15	0.92	0.90 / 1.15	0.98
WikiCS	APS	0.92 / 3.70	0.94	0.93 / 3.62	0.95
	TPS	0.93 / 3.52	0.94	0.93 / 3.52	0.95

Table 4: AEGIS-Conformal vs. randomized smoothing (Cora, $n=200$, 10 seeds, $\alpha=0.1$, APS). **frob**: matched Frobenius ball ($\sigma = \varepsilon / \sqrt{2|E|}$); **per_edge**: larger per-coordinate ball ($\sigma = \varepsilon$), favorable to smoothing. cert: fraction certified; wall-clock at $M=10^4$ samples (AEGIS-Conformal is zero-sample).

method	match	ε	cov	set	cert	wall@10 ⁴
AEGIS-Conformal	–	0.01	0.90	1.36	–	~ 1 s
RandSmoothing	frob	0.01	1.00	7.00	0.00	7,600 s
RandSmoothing	per_edge	0.01	0.95	1.26	0.96	15,000 s
AEGIS-Conformal	–	0.05	0.90	1.05	–	~ 1 s
RandSmoothing	frob	0.05	1.00	7.00	0.00	10,900 s
RandSmoothing	per_edge	0.05	0.99	2.39	0.77	36,300 s

Table 5: The $\sigma_1(S_c)$ penalty trades clean accuracy for certified robustness (Cora, $c=0.9$ IGNN, 10 seeds). Cert. frac peaks at $\lambda=3 \times 10^{-3}$ before margin collapse.

λ	Acc	$\sigma_1(S_c)$	Cert. frac	Attack dmg
0	.781 $\pm$.018	319 $\pm$ 59	.40 $\pm$.08	27.8 $\pm$ 6.3
3×10^{-4}	.739 $\pm$.008	32.6 $\pm$ 1.7	.82 $\pm$.03	3.1 $\pm$ 0.2
1×10^{-3}	.690 $\pm$.007	10.7 $\pm$ 0.6	.89 $\pm$.02	1.1 $\pm$ 0.1
3×10^{-3}	.619 $\pm$.006	3.9 $\pm$ 0.2	.92 $\pm$.02	0.4 $\pm$ 0.0
1×10^{-2}	.564 $\pm$.006	0.8 $\pm$ 0.1	.86 $\pm$.04	0.1 $\pm$ 0.0

(section D.3).

5.3 Defend: A $\sigma_1(S_c)$ Robustness Knob

The operator that finds the worst attack also tunes the defense. Penalizing $\sigma_1(S_c)$ during training drives the dominant sensitivity down, and with it the worst-case attack damage falls while the certified fraction rises (table 5). At the operating point $\lambda=3 \times 10^{-4}$ this cuts σ_1 by $\sim 10 \times$ ($319 \pm 59 \rightarrow 32.6 \pm 1.7$) and lifts the certified fraction to 0.82 ± 0.03 , at a cost of roughly 4 accuracy points. The gain is genuinely *certified* rather than empirical: at $\varepsilon \leq 0.1$ few nodes flip for either model (the delocalized vulnerability noted above), so the penalty buys a larger certified margin, not better small-budget accuracy. Attack and defense thus couple through the one operator: across 10 seeds the margin-controlled attack magnitude and the certified radius are anticorrelated (-0.65 ± 0.12 , negative in 10/10 seeds, $p \approx 10^{-160}$).

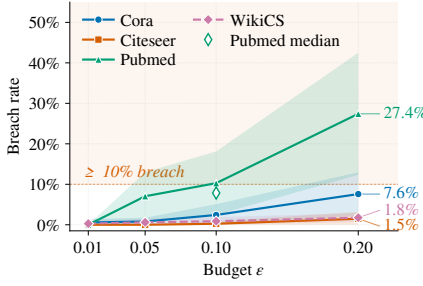


Figure 3: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ϵ (10 seeds; mean, 95% CI bands).

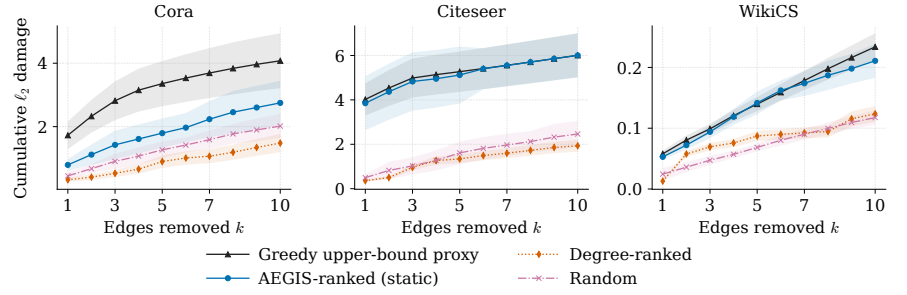


Figure 4: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds). AEGIS matches the label-aware Greedy attacker on Citeseer/WikiCS and recovers 54–67% of its Cora damage at $k=5-10$ with no discrete-label access; degree-ranking stays below random.

Kendall τ : continuous $\rightarrow$ discrete transfer

	Cora	Citeseer	Pubmed	WikiCS	Amazon Photo	Amazon Fraud
IGNN	+0.98	+0.98	+0.99	+1.00	+1.00	+0.99
GCN-2	+0.99	+0.99	+1.00	+0.99	+0.99	+0.98
GCN-4	+0.91	+0.91	+0.95	+1.00	+1.00	+1.00
GIN-2	+0.91	+0.91	+0.96	+0.99	+0.96	+0.95
GAT-2 [†]	+0.32	+0.50	OOM	OOM	+0.23	OOM
SAGE-2	+0.98	+0.98	+0.99	+1.00	+0.99	+0.98
APNP	+0.98	+0.97	+0.99	+1.00	+1.00	+1.00

Color scale for Kendall τ : 0 (red) to 1 (blue). Cross-hatch indicates GPU OOM.

[†] GAT-2: 2-layer attention; OOM on Pubmed, WikiCS, Amazon Fraud.

Figure 5: Continuous-to-discrete transfer τ (10-seed means; per-cell sd 0.0003–0.041). Cross-hatch: GPU OOM. GAT-2[†] is lower under weighting (it transfers better unweighted; see section D.6).

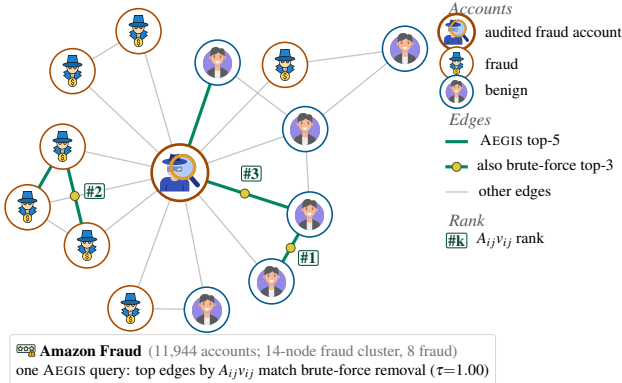


Figure 6: AEGIS auditing an IGNN fraud detector on an Amazon Fraud cluster (Dou et al. 2020). Icons mark fraud / benign accounts; the audited account sits at the hub. Green edges are the top-5 adversarial fault lines by the edge-weighted score $A_{ij}v_{ij}$; yellow dots mark the three also in the brute-force single-edge-removal damage top-3.

6 Case Study: Auditing a Fraud Detector

GNN fraud detectors are a prime adversarial target: a flagged account can slip past detection by perturbing a few behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits such a

detector directly, with no labels and no retraining, reading off from a *single* S_c query which edges most threaten its predictions. Figure 6 shows the audit on an Amazon Fraud cluster: the edge-weighted ranking $A_{ij}v_{ij}$ (Proposition 3) pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking ($\tau=1.0$ on this cluster) at the cost of one query rather than $|E|$ reconvergences. The audit is *non-circular*: the ranking transfers across the full fraud graph (fig. 5, where uniform edge weights are least informative, so v_{ij} adds the most rank information, with $\Delta\tau$ up to +0.65), and the SVD direction transfers from a separately trained surrogate (section 5.1), confirming that v_{ij} is model-intrinsic rather than a gradient artifact. The full walk-through is in section E.

7 Related Work

Each of three threads supplies at most one of AEGIS’s diagnostics. **Structural attacks**, among them Nettack (Zügner, Akbarnejad, and Günnemann 2018), Mettack (Zügner and Günnemann 2019a), GR-BCD/PR-BCD (Geisler et al. 2021) and variants (Wu et al. 2019; Xu et al. 2019a), optimise surrogate gradients to flip predictions but yield no analytic sensitive direction and no per-node radius; sober-look studies (Mujkanovic et al. 2022; Gosch et al. 2023) flag the evaluation pitfalls we adopt (section D.5).

Certified robustness and defense. Convex relaxations (Zügner and Günnemann 2019b), smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023), collective (Schuchardt et al. 2021) and majority-vote (Li and Wang 2025) certificates, and weight-norm bounds (Abbahaddou et al. 2024) certify per-node robustness but do not name vulnerability-driving edges; empirical defenses (Zhu et al. 2019; Jin et al. 2020; Zhang and Zitnik 2020) harden models without surfacing per-edge sensitivity. Graph conformal prediction (Zargarbashi, Antonelli, and Bojchevski 2023) certifies a *fixed* graph, whereas AEGIS-Conformal covers the adversarial ϵ -ball via the S_c score-shift bound (section 4.1).

Implicit networks and influence functions. DEQs (Bai, Kolter, and Koltun 2019), IGNN (Gu et al. 2020), monotone equilibria (Winston and Kolter 2020), and well-posedness analyses (El Ghaoui et al. 2021) address *input* sensitivity

$\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$, applying the Koh–Liang influence lineage (Koh and Liang 2017) and implicit differentiation (Lorraine, Vicol, and Duvenaud 2020; Gould, Hartley, and Campbell 2021) to a *structural-parameter* perturbation via P_c rather than to a loss. GNN explainers (Ying et al. 2019) rank edges for *fidelity*, not vulnerability; matrix perturbation theory (Stewart and Sun 1990; Trefethen and Embree 2005) supplies the pseudospectral toolkit; and curvature/over-squashing (Topping et al. 2022; Di Giovanni et al. 2023) and stability bounds (Gama, Bruna, and Ribeiro 2020; Kenlay, Thanou, and Dong 2021) tie edge structure to sensitivity only in aggregate, whereas AEGIS localises it per edge and node.

8 Conclusion

From one matrix-free object, S_c , AEGIS *audits*, *certifies*, and *defends* any GNN with continuous edge-weight message passing, adding for contractive implicit models the safety boundary of Theorem 1. Attacking, certifying, and defending, once three separate tools, become three readings of a single operator that is computed in one pass and coupled by construction.

Limitations. The radii r_v are first-order thresholds, and AEGIS-Conformal currently runs densely at $N=200$. Standard GAT and binary-mask architectures fall outside the framework (section 5.1), the $\varepsilon_{\text{crit}}$ track costs $\sim 6\%$ accuracy, and insertion attacks are scoped out (section 2). Finally, AEGIS audits vulnerability, not physics: a contractive surrogate cannot represent voltage collapse, so it complements rather than replaces power-flow contingency screening (Donon et al. 2019; Nakiganda et al. 2023; Varbella, Gjorgiev, and Sansavini 2024).

Disclosure. We propose a 90-day coordinated-notification window: the diagnostic path (r_v, v_{ij}) releases unconditionally, while attack-direction reconstruction (algorithm 1) is gated behind ethics review, since it alone yields the damaging δA^* (Proposition 1).

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A Proofs for the Adversarial Equilibrium Theory

This appendix collects the proof bodies for the results stated in section 4. We restate each result in brief for self-containment.

A.1 Proof of Theorem 1

Recall Theorem 1 characterises three regimes of the perturbation budget ε for an IGNN-class operator under (A1)–(A3), with critical budget $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$.

Proof. (a). Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\|\delta\hat{A}\|$ (the boundary set $\{\hat{A} : z^*(\hat{A}) \text{ lies on a region face}\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels (Bolte and Pauwels 2021) yields a conservative IFT on each region. With $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \|\hat{A}\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta\hat{A}) + O(\|\delta\hat{A}\|^2), \quad (9)$$

with $J_A = \partial F / \partial \text{vec}(A)$ and Neumann $\|(I - J_z)^{-1}\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). For any M , $(I - M)^{-1}$ has eigenvalues $(1 - \lambda_i(M))^{-1}$, hence $\|(I - M)^{-1}\|_2 \geq \rho((I - M)^{-1}) = 1/\min_i |1 - \lambda_i(M)|$ *unconditionally*. Along the worst-case direction $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$, so $1 - \|J'_z\|_2 \geq \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$ and, since $\rho \leq \|\cdot\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ certifies $\|J'_z\|_2 < 1$. The certificate is one-sided: $\|J'_z\|_2 \rightarrow 1$ need not drive $\min_i |1 - \lambda_i| \rightarrow 0$ (e.g. $M = \text{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\|(I - M)^{-1}\|_2 = 1$). When J'_z is normal with dominant real-positive eigenvalue $\lambda^* = \|J'_z\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1 - \lambda_i| = 1 - \lambda^* = \|W\|_2 (\varepsilon_{\text{crit}} - \varepsilon)$ yields the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1 - \lambda_i|$ away from 0 (the same $\text{diag}(-s, 0)$), so $\varepsilon_{\text{crit}}$ only lower-bounds the threshold, slack controlled by η (Observation 1). For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. $\square$

A.2 Graph-Independent Nonnormality Bound

Observation 1 (Graph-Independent Nonnormality Bound, all-active case). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$, independent of $\hat{A}$.*

Remark 1 (Empirical extension to general ReLU patterns). *The proof requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section D.5), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W .*

Proof. $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ (Stewart and Sun 1990); symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}}) = 1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$ yields $\eta \leq \kappa(V_W)$. $\square$

A.3 Proof of Proposition 2

Proof. By Theorem 1(a), $\Delta z_v^* \approx S_v \text{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v} - f_c)| \leq \|(W_{y_v} - W_c)S_v\|_2 \|\delta A\|_F$ by Cauchy–Schwarz. The prediction survives while every margin stays positive; taking the binding minimum over competitors gives

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)S_v\|_2}.$$

Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii. $\square$

A.4 Continuous-to-Discrete Ranking Transfer

Proposition 3 (Continuous-to-Discrete Ranking Transfer).

Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i, j) \in E$, $v_k = \| [S_c]_{:,k} \|_2$ the continuous vulnerability score, and $d_k = \| z^*(A) - z^*(A \setminus k) \|_2$ the discrete damage from fixed-normalization removal of edge k . Assume all single-edge removals are subcritical: $\sqrt{2} \max_k w_k < \varepsilon_{\text{crit}}$.

(a) **First-order bridge.** The discrete damage satisfies

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \quad (10)$$

where $L_J \leq \|W\|_2^2 \|z^*\|$ is the second-order curvature constant of $z^*(A)$, finite at the fixed point.

(b) **Ranking preservation (pairwise-sufficient).** For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (11)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2} (v_{k_1} - v_{k_2}) \cdot (1 - \kappa)^2 / L_J$.

Empirically $|R_k|$ is 2–10 $\times$ smaller than $L_J w_k^2$, the constrained S_c tightens the bound from 0.31 to 1.00, and the edge-weighted score $w_k v_k$ reaches $\tau = +0.996$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section 5.1).

Proof. (a). We model deletion as *fixed-normalization* masking, $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $O(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section 5.1.) For the remainder, the IFT gives $\partial z^* / \partial \text{vec}(A) = (I - J_z)^{-1} J_A$, and the dominant curvature term $(I - J_z)^{-1} (\partial J_z / \partial \text{vec}(A)) (I - J_z)^{-1} J_A$ carries two resolvents $(1 - \kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z / \partial \text{vec}(A) \propto W$ and $J_A \propto z^* W^\top$), so $L_J \leq \|W\|_2^2 \|z^*\|$, finite since $\|z^*\| \leq \|X_{\text{proj}}\| / (1 - \kappa)$ (the κ -contraction of F at the fixed point, A3), giving $\|\partial^2 z^* / \partial \text{vec}(A)^2\|_2 \leq (1 - \kappa)^{-2} L_J$. The path meets finitely many ReLU regions (simultaneous crossings non-generic, measure zero); summing the second-order remainder over these regions, with the conservative IFT (Bolte and Pauwels 2021) on each, gives $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$. (b). From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1 - \kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (11) ensures the first-order gap dominates. $\square$

A.5 Structural Sensitivity for K -Layer Explicit GNNs

Proposition 4 (Structural Sensitivity for K -Layer GNNs).

Let $Z_K = g_K \circ \dots \circ g_1(X, A)$ be a K -layer GNN where

each layer g_l is differentiable with respect to both its input and A . Define the unrolled sensitivity matrix

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (12)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$. Then:

(a) The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with

$$\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2. \quad (13)$$

(b) The constrained construction S_c and per-node radius r_v (Proposition 2) apply identically with S_K in place of S .

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 (1 - \kappa^K) / (1 - \kappa)$ converges to the IGNN bound as $K \rightarrow \infty$. Explicit GNNs thus inherit the full pipeline (empirical tightness 0.99–1.02 across the suite, section 5.1); only the closed-form $\varepsilon_{\text{crit}}$ is restricted to the contractive subclass.

B Constant-Factor Two-Sided Boundary Theorem

This appendix gives the full statement and proof of Theorem 2 (the constant-factor two-sided boundary cited in section 4), together with the empirical nonlinear extension (Observation O1). The lower side (i) is the sound certificate sold by the paper; the upper side (ii)–(iii) is proved for the all-active operator (A4).

Notation and standing assumptions. $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ is the symmetric normalized adjacency; W the tied weight; ϕ the elementwise activation; z^* the equilibrium; the all-active Jacobian is $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$, $\kappa := \|J_z\|_2$, $\hat{A}' = \hat{A} + \delta \hat{A}$, $J'_z = \text{diag}(\phi')(\hat{A}' \otimes W)$.

Definition 1 (Operator and certified regime). Let $F_\theta(Z, \hat{A}) = \phi(\hat{A} Z W^\top + X_{\text{proj}})$ with:

- (A1) ϕ is 1-Lipschitz (ReLU or any such), so $\|\text{diag}(\phi')\|_2 \leq 1$ everywhere; nonsmoothness at 0 is handled by the conservative IFT (Bolte and Pauwels 2021).
- (A2) $\|W\|_2 \leq c$ via spectral normalization (controls the numerator of g_W only; see the a-posteriori remark below).
- (A3) The trained model is contractive: $\kappa = \|J_z\|_2 < 1$, reported and audited post-training ($\kappa \in [0.14, 0.59]$ across our suite). Equivalently $\|\hat{A}\|_2 \|W\|_2 < 1$.
- (A4) All-active. On the operating set $\phi' \equiv 1$, so $J_z = \hat{A} \otimes W$. The upper bound (ii)–(iii) of Theorem 2 is proved for this operator; the lower bound (i) does not use (A4).

Audited (a-posteriori) gap quantities. Define $g_W := \|W\|_2 / \rho(W) \geq 1$ and $g_A := \|\hat{A}\|_2 / \rho(\hat{A}) = 1$ ($\hat{A}$ symmetric). g_W is the nonnormality of W , stated as an audited per-model constant, not a quantity controlled a priori by training: spectral normalization (A2) bounds $\|W\|_2$ but not $\rho(W)$

away from 0, so in principle $\rho(W) \rightarrow 0$ at fixed $\|W\|_2$ sends $g_W \rightarrow \infty$. We *measure* g_W on each trained model ($g_W \in [1.19, 2.47]$ in our suite, with the a-posteriori certificate $g_W \leq \kappa_2(V_W)$, the eigenvector conditioning of W) and report C as a function of that value. Every constant in Theorem 2 is computable post-training from $\rho(W)$, $\|W\|_2$, $\|\hat{A}\|_2$ and the edge set.

Two budgets and the boundary.

$$\varepsilon_{\text{crit}} := \frac{1 - \kappa}{\|W\|_2} = \frac{1}{\|W\|_2} - \|\hat{A}\|_2 \quad (\text{norm certificate}),$$

$$\varepsilon_{\text{spec}} := \frac{1}{\rho(W)} - \rho(\hat{A}) \quad (\text{all-active spectral break}).$$

$\varepsilon_{\text{br}}^{\text{all}} :=$ the infimal $\|\delta\hat{A}\|_F$ over feasible (symmetric, edge-supported) perturbations at which the *all-active* operator loses contraction, $\rho(\hat{A}' \otimes W) \geq 1$; the superscript marks that the upper-bounded object is the all-active linearization, not the deployed ReLU model’s stability boundary.

Edge-support alignment. Let P_E be the orthogonal projection onto the symmetric, edge-supported subspace $\mathcal{S}_E := \{M = M^\top : M_{ij} = 0 \text{ if } (i, j) \notin E \cup \{(i, i)\}\}$. With u_1 a unit top eigenvector of $\hat{A}$, define $g_E := P_E(u_1 u_1^\top)$, $\sigma_E := \|g_E\|_F \in [0, 1]$, and (when $\sigma_E > 0$) the unit feasible direction $B := g_E / \sigma_E$. Then $\beta := \langle u_1, B u_1 \rangle = \sigma_E^2 / \sigma_E = \sigma_E \in (0, 1]$, using $\langle u_1, P_E(u_1 u_1^\top) u_1 \rangle = \|g_E\|_F^2 = \sigma_E^2$ (P_E an orthogonal projection, $u_1 u_1^\top$ symmetric). Thus the alignment β equals the edge-supported Frobenius mass $\sigma_E = \|P_E(u_1 u_1^\top)\|_F$ of the rank-one critical direction, strictly positive whenever the top Perron mode places mass on the edge set (always true for a connected graph, $u_1 > 0$ entry-wise, $E \neq \emptyset$), making the hypothesis $\beta > 0$ an observable rather than an assumption.

Theorem 3 (Constant-factor two-sided bracket for the all-active IGNN contraction boundary, full statement). *Let Definition 1 hold with $\varepsilon_{\text{crit}} > 0$ (certified regime) and $\hat{A}$ symmetric, and let $\beta = \sigma_E \in (0, 1]$ be the edge-supported mass of the critical mode. Then:*

- (i) **Lower side: sound certificate (any 1-Lipschitz ϕ ; no (A4)).** For every feasible $\delta\hat{A}$ with $\|\delta\hat{A}\|_F \leq \varepsilon < \varepsilon_{\text{crit}}$, the perturbed operator $F_\theta(\cdot, \hat{A} + \delta\hat{A})$ is an $\|\cdot\|_2$ -contraction ($\|J'_z\|_2 < 1$) with a unique equilibrium and $\rho(J'_z) < 1$. Hence $\varepsilon_{\text{br}} \geq \varepsilon_{\text{crit}}$; in particular $\varepsilon_{\text{br}}^{\text{all}} \geq \varepsilon_{\text{crit}}$.
- (ii) **Upper side: all-active matching attack (requires (A4)).** There is a feasible rank-one $\delta\hat{A}^* = (\varepsilon_{\text{spec}}/\beta) B \in \mathcal{S}_E$ with $\|\delta\hat{A}^*\|_F = \varepsilon_{\text{spec}}/\beta$ that drives the all-active spectral radius to 1; hence $\varepsilon_{\text{br}}^{\text{all}} \leq \varepsilon_{\text{spec}}/\beta$. In the unconstrained-symmetric threat model ($P_E = I$, $\beta = 1$) the construction is exact: $\varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$ with extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$.
- (iii) **Constant-factor enclosure.** $\varepsilon_{\text{spec}} \leq C \varepsilon_{\text{crit}}$ with the ex-

plicit, dimension-free, a-posteriori-computable constant

$$C := g_W \frac{1 + \kappa}{1 - \kappa} = \frac{\|W\|_2}{\rho(W)} \cdot \frac{1 + \|\hat{A}\|_2 \|W\|_2}{1 - \|\hat{A}\|_2 \|W\|_2}.$$

Combining with (i)–(ii), the all-active contraction boundary obeys

$$\boxed{\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}}^{\text{all}} \leq \frac{C}{\beta} \varepsilon_{\text{crit}}}. \quad (14)$$

The bracket is non-vacuous whenever $\beta = \sigma_E > 0$ (every connected graph); on the suite $\beta \approx 0.62$, giving $C/\beta \lesssim 16$. The β -free form $\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}}^{\text{all}} \leq C \varepsilon_{\text{crit}}$ holds exactly for the unconstrained-symmetric model ($\beta = 1$); we report C/β as the certified edge-feasible constant and C as the $\beta = 1$ specialization.

- (iv) **Exact boundary (rate-sharp regime), separated equality conditions.** (a) Endpoints coincide: $\varepsilon_{\text{crit}} = \varepsilon_{\text{spec}} \iff g_W = 1$ (W normal); κ, β irrelevant. (b) All three coincide: $\varepsilon_{\text{crit}} = \varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}} \iff g_W = 1$ and $\beta = 1$ (the exact extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$ is edge-feasible). (c) Slack constant equals one: $C/\beta = 1 \iff g_W = 1, \beta = 1$, and $\kappa \rightarrow 0^+$ (kills the $\frac{1+\kappa}{1-\kappa}$ inflation).

This is a constant-factor (dimension-free, $O(1)$) two-sided characterisation, rate-sharp in the normal/edge-aligned regime, with enclosure constant C/β reaching ~ 10 – $16\times$ on our suite (random-instance $\varepsilon_{\text{spec}}/\varepsilon_{\text{crit}} \in [1.02, 18.2]$, median 2.25); it is not “tight” in the matching-bound sense except in the corner (iv)(b). Two structural identities underlie it (both numerically exact over 4,000–6,000 random certified-regime instances): the *all-active spectral law* $\rho(J'_z) = \rho(\hat{A}')\rho(W)$, $\|J'_z\|_2 = \|\hat{A}'\|_2 \|W\|_2$ (A4 only); and the *additive gap identity* $\varepsilon_{\text{spec}} - \varepsilon_{\text{crit}} = (\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) + (\|\hat{A}\|_2 - \rho(\hat{A})) \geq 0$, the slack supplied entirely by W ’s non-normality when $\hat{A}$ is symmetric (verified to 2.7×10^{-15}).

Proof. Setup. Feasible perturbations lie in $\mathcal{S}_E$ with $\|\delta\hat{A}\|_F = \varepsilon$. We use Weyl, $\rho(\hat{A}') \leq \rho(\hat{A}) + \|\delta\hat{A}\|_2$, $\|\hat{A}'\|_2 \leq \|\hat{A}\|_2 + \|\delta\hat{A}\|_2$, and $\|\delta\hat{A}\|_2 \leq \|\delta\hat{A}\|_F = \varepsilon$. Since $\hat{A}$ and the extremal $\delta\hat{A}$ are symmetric, $\rho(\hat{A}') = \|\hat{A}'\|_2$ and Weyl holds with equality along an aligned rank-one mode.

(i) **Lower side, fully nonlinear.** By (A1) $\|\text{diag}(\phi')\|_2 \leq 1$; the spectral norm is sub-multiplicative and Kronecker-multiplicative, so $\|J'_z\|_2 \leq \|\text{diag}(\phi')\|_2 \|\hat{A}' \otimes W\|_2 \leq \|\hat{A}'\|_2 \|W\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$. For $\varepsilon < \varepsilon_{\text{crit}} = \frac{1}{\|W\|_2} - \|\hat{A}\|_2$ this is < 1 . A spectral-norm contraction has a unique fixed point (Banach) and $\rho(J'_z) \leq \|J'_z\|_2 < 1$, so no feasible $\varepsilon < \varepsilon_{\text{crit}}$ breaks contraction of the deployed model: $\varepsilon_{\text{br}} \geq \varepsilon_{\text{crit}}$. No (A4) used.

Additive gap. $\varepsilon_{\text{spec}} - \varepsilon_{\text{crit}} = (\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) + (\|\hat{A}\|_2 - \rho(\hat{A})) \geq 0$ using $\rho \leq \|\cdot\|_2$; with $\hat{A}$ symmetric the second bracket is 0, so $\varepsilon_{\text{crit}} \leq \varepsilon_{\text{spec}}$.

(ii) Upper side, all-active matching attack (A4). Let u_1 be a unit top eigenvector of symmetric $\hat{A}$. With the rank-one symmetric direction $t u_1 u_1^\top$ ($\|u_1 u_1^\top\|_F = 1$), $\rho(\hat{A} + t u_1 u_1^\top) = \rho(\hat{A}) + t$ exactly. Under (A4), $\rho(J'_z) = (\rho(\hat{A}) + t)\rho(W)$ reaches 1 at $t = \varepsilon_{\text{spec}} = \frac{1}{\rho(W)} - \rho(\hat{A})$, so the unconstrained all-active break budget is exactly $\varepsilon_{\text{spec}}$ with extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$. For edge feasibility, project to $\mathcal{S}_E$ via $B = P_E(u_1 u_1^\top)/\sigma_E$, so $\beta = \sigma_E = u_1^\top B u_1 > 0$. The map $t \mapsto \lambda_{\max}(\hat{A} + tB) = \max_{\|v\|=1} v^\top(\hat{A} + tB)v$ is a pointwise maximum of affine functions, hence *convex*, so it lies above its tangent $\rho(\hat{A}) + \beta t$ at $t = 0$: $\rho(\hat{A} + tB) \geq \rho(\hat{A}) + \beta t$ for all $t \geq 0$ (numerically: 0/42000 violations of this lower bound, vs. 42000/42000 for a concave version). Choosing $\delta \hat{A}^* = (\varepsilon_{\text{spec}}/\beta)B \in \mathcal{S}_E$ drives $\rho(\hat{A}') \geq \rho(\hat{A}) + \varepsilon_{\text{spec}} = \frac{1}{\rho(W)}$, so $\rho(J'_z) \geq 1$ and $\varepsilon_{\text{br}}^{\text{all}} \leq \|\delta \hat{A}^*\|_F = \varepsilon_{\text{spec}}/\beta$; exact when $\beta = 1$.

(iii) Closing the bracket. Dropping $-\rho(\hat{A}) \leq 0$ and using $\rho(\hat{A}) = \|\hat{A}\|_2$, $\varepsilon_{\text{spec}} \leq \frac{1}{\rho(W)} = g_W \frac{1}{\|W\|_2} = g_W(\varepsilon_{\text{crit}} + \|\hat{A}\|_2)$. Since $\|\hat{A}\|_2 = \kappa/\|W\|_2 = \kappa(\varepsilon_{\text{crit}} + \|\hat{A}\|_2)$ gives $\|\hat{A}\|_2 = \frac{\kappa}{1-\kappa}\varepsilon_{\text{crit}}$, we get $\varepsilon_{\text{spec}} \leq g_W \frac{1}{1-\kappa}\varepsilon_{\text{crit}} \leq g_W \frac{1+\kappa}{1-\kappa}\varepsilon_{\text{crit}} = C \varepsilon_{\text{crit}}$ (verified, zero violations over 6,000 instances). With (i) and (ii) the boxed bracket (14) follows.

(iv) Equality conditions. (a) The gap $(\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) \geq 0$ is zero iff $g_W = 1$; κ, β do not appear. (b) Add edge-feasibility of $\varepsilon_{\text{spec}} u_1 u_1^\top$, i.e. $\beta = 1$; then $\varepsilon_{\text{crit}} = \varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$. (c) Additionally $\frac{1+\kappa}{1-\kappa} = 1$ requires $\kappa \rightarrow 0^+$. $\square$

This recovers and sharpens Theorem 1: the old result is the lower side (i); (ii) supplies the matching all-active upper side; (iv) separates the three equality conditions.

Remark 2 (Observation O1: nonlinear break budget and the active fraction; empirical / conjecture, not part of Theorem 2). *The deployed model is ReLU, not all-active; its measured nonlinear break budget $\varepsilon_{\text{reach}}$ exceeds the all-active $\varepsilon_{\text{spec}}$. Measured (reachability over the 10 preferred seeds, 50-node ego subgraph; $\|\hat{A}\|_F = 1.77$): at $\kappa_0=0.5$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}}=1.41\pm 0.12$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}=2.17\pm 0.18$, *resolvent* $\gamma=1.06\pm 0.09$; at $\kappa_0=0.9$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}}=1.51\pm 0.18$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}=8.72\pm 1.03$, $\gamma=1.02\pm 0.02$. Empirically $\varepsilon_{\text{reach}} \approx (1/a)\varepsilon_{\text{spec}}$ with a the ReLU active fraction ($a := \frac{1}{Nd} \mathbb{E} \|\text{diag}(\phi')\|_0 \in (0, 1]$; with $a \approx 0.6$, $1/a \approx 1.5$ – 1.7 , matching $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}} \in [1.4, 1.5]$. Composing with the bracket gives the observed envelope $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} \lesssim C/a$, spanning roughly $[2, 10]$ and matching the measured $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} \in [2.2, 8.7]$.*

This is a conjecture; two independent proof gaps remain open. (I) Masked-operator spectral scaling (unproved): $\varepsilon_{\text{reach}} \approx \varepsilon_{\text{spec}}/a$ assumes the dominant eigenvalue of the

Table 6: Reconciliation of the robustness constants, separating the three “ $\times$ ” ratios that recur in the text: they are *distinct* quantities, not estimates of one number. The operating margin measures how far the trained model sits inside $\rho=1$; the empirical conservatism is how far $\varepsilon_{\text{crit}}$ under-states the true nonlinear break; the proven bracket is the worst-case enclosure constant.

quantity	meaning	source
$\kappa = \ J_z\ _2$	trained contractivity (< 1)	table 1
$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\ W\ _2}$	norm certificate (safe radius)	Theorem 1
$\varepsilon_{\text{spec}}$	all-active spectral break budget	Theorem 2
$2\text{--}4\times$	operating ρ -margin to $\rho=1$	section D.4
$2\text{--}9\times$	empirical $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}$ (10 seeds)	Remark 2
$10\text{--}16\times$	proven bracket constant C/β	Theorem 2
$\gamma \approx 1.02$	resolvent critical exponent	Remark 2
$g_W \in [1.19, 2.47]$	non-normality $\ W\ _2/\rho(W)$	Theorem 2

masked Jacobian $D_a(\hat{A}' \otimes W)$ scales as $\approx a \rho(\hat{A}')\rho(W)$, but no such lemma holds in general (the active set correlates with the equilibrium and with u_1); this is a mean-field heuristic calibrated to two operating points. (2) Linear-to-nonlinear bifurcation (empirical only): the jump from “a linearized-Jacobian eigenvalue reaches 1” to “the true nonlinear fixed point destabilizes” is supported here only empirically across the 10 preferred seeds (a clean simple-pole divergence, $\gamma \approx 1.02$ – 1.06). We present $\varepsilon_{\text{reach}} \approx (1/a)\varepsilon_{\text{spec}}$ as an empirical regularity organized by the same two knobs (g_W, a), not a proven extension. Practical caveat: $\varepsilon_{\text{reach}}$ is 60–128% of $\|\hat{A}\|_F$, i.e. $5\text{--}11\times$ the largest realistic budget ($\varepsilon \leq 0.2$), so realistic-budget robustness is governed by the first-order $\sigma_1(S_c)$, not by proximity to criticality.

C Derivation of the AEGIS-Conformal Bound

This appendix supplies the derivation behind eq. (8): the worst-case conformity-score-shift bound $L_1^c \varepsilon + C_v \varepsilon^2$ and the robust-coverage guarantee. The argument has three parts. Definition 2 fixes the two conformity scores and the two constants. Lemma 1 proves the per-node worst-case score-shift bound from the first-order sensitivity of Theorem 1(a) (linear term, via Cauchy–Schwarz) and the second-order curvature remainder of Proposition 3 (quadratic term). Theorem 4 inflates the split-conformal threshold by this shift and reduces to the binary split-conformal certificate of Zargarbashi, Antonelli, and Bojchevski (2023), valid over the entire ε -ball under a stated exchangeability condition. Throughout, (A1)–(A3) are the standing assumptions of Theorem 1 and we operate in the certified regime $\varepsilon < \varepsilon_{\text{crit}}$.

Setup and notation. Fix a target node v with true label $y_v \in [C]$. Let $f(z^*) = Wz_v^* + b \in \mathbb{R}^C$ be the node’s pre-softmax logits read off the equilibrium $z^* = z^*(\hat{A})$ by the linear head W (rows $W_1, \dots, W_C$), and $\pi = \text{softmax}(f) \in \Delta^{C-1}$ the class probabilities. For a feasible structural perturbation $\delta \hat{A} \in \mathcal{S}_E$ (symmetric, edge-supported; section 2) with $\|\delta \hat{A}\|_F \leq \varepsilon$, write $z^{*'} = z^*(\hat{A} + \delta \hat{A})$, $\Delta z^* = z^{*'} - z^*$, and $f' = Wz^{*'} + b$, $\pi' = \text{softmax}(f')$. Let $S_{c,v} \in \mathbb{R}^{d \times |E|}$ be the block-rows of the constrained sensitivity matrix S_c at

node v (Proposition 2), so by Theorem 1(a),

$$\begin{aligned} \Delta z_v^* &= S_{c,v} \delta + R_v(\delta \hat{A}), \\ \delta &:= (\text{edge-basis coords of } \delta \hat{A}), \|\delta\|_2 = \|\delta \hat{A}\|_F \leq \varepsilon, \end{aligned} \quad (15)$$

where R_v is the IFT second-order remainder bounded in Proposition 3. (The edge basis $\{b_k\}$ is orthonormal, $\|b_k\|_F = 1$, so the coordinate map is an isometry and $\|\delta\|_2 = \|\delta \hat{A}\|_F$; this is exactly the normalization that makes $\sigma_1(S_c)$ consistent with the Frobenius budget, Proposition 1.)

Definition 2 (Conformity scores and the two AEGIS constants). *For node v and candidate label $r \in [C]$ define the two conformity scores used by AEGIS, TPS (Sadinle, Lei, and Wasserman 2019) and APS (Romano, Sesia, and Candès 2020) (higher = more conforming; the prediction set is $C(v) = \{r : \text{score}_r(v) \geq \hat{q}\}$):*

$$\text{TPS: } \text{score}_r^{\text{TPS}}(v) = \pi_r, \quad (16)$$

$$\text{APS: } \text{score}_r^{\text{APS}}(v) = 1 - (\rho_r + u_v \pi_r), \quad (17)$$

$$\rho_r := \sum_{c: \pi_c > \pi_r} \pi_c, \quad (18)$$

with $u_v \sim \text{Unif}[0, 1]$ a per-node tie-break drawn once and shared between calibration and test (so it cancels under exchangeability). For each competitor $c \neq r$ let $g_c := f_r - f_c$ be the logit margin of r against c . Define

$$L_1^c := L_{1,v}^{(c)} := \|(W_r - W_c) S_{c,v}\|_2, \quad (19)$$

$$C_v := \|W_r - W_c\|_2 \cdot \frac{L_{J,v}}{2(1 - \kappa)^2}, \quad (20)$$

where $L_{J,v} = \|\partial J_z / \partial \text{vec}(A)\| \cdot (\text{equilibrium norm}) \leq \|W\|_2^2 \|z^*\|$ is the per-node IFT curvature constant of Proposition 3(a) and $\kappa = \|J_z\|_2 < 1$ (A3). Equation (19) is the first-order margin sensitivity; (20) is the curvature constant $C = L_J / (2(1 - \kappa)^2)$ of Proposition 3 scaled by the readout gap $\|W_r - W_c\|_2$. The aggregate node constant reported in eq. (8) is the competitor-worst value $L_1^c = \max_{c \neq r} \|(W_r - W_c) S_{c,v}\|_2$ (and C_v likewise with the worst $\|W_r - W_c\|_2$).

Remark 3 (Why the gradient is the readout gap, not $\nabla_z \text{score}$). For TPS, $\text{score}_r = \pi_r$ and $\nabla_z \text{score}_r = \pi_r (W_r - \sum_c \pi_c W_c)$ is a genuine softmax-Jacobian row; $\|(\nabla_z \text{score}_r)^\top S_{c,v}\|$ is then a valid first-order constant. The implemented constant (19) instead uses the logit-margin rows $W_r - W_c$, one per competitor. This is the correct object for two reasons. (i) Both scores are monotone functions of the margins $\{g_c\}_{c \neq r}$ alone (softmax depends on logits only through differences), so controlling every g_c controls π_r , ρ_r , and hence both scores (Lemma 1). (ii) The margin form needs no bound on softmax curvature: it yields a sound (one-sided) score drop without the 1/4 softmax-Lipschitz slack. We therefore write $L_1^c := \max_{c \neq y_v} \|(W_{y_v} - W_c) S_{c,v}\|_2$ rather than the $(\nabla_z \text{score}_v)^\top S_{c,v}$ form, which is literally correct only for TPS.

Lemma 1 (Worst-case conformity-score shift). *Assume (A1)–(A3) and $\varepsilon < \varepsilon_{\text{crit}}$, and that the readout $f = W z_v^* + b$ is affine in z_v^* (a linear classification head). Then for every feasible $\delta \hat{A}$ with $\|\delta \hat{A}\|_F \leq \varepsilon$ and every label $r \in [C]$,*

$$\begin{aligned} |g_c(\hat{A} + \delta \hat{A}) - g_c(\hat{A})| &\leq L_{1,v}^{(c)} \varepsilon + C_v \varepsilon^2 \\ &\text{for every competitor } c \neq r, \end{aligned} \quad (21)$$

and consequently the worst-case drop of the conformity score of label r obeys

$$\begin{aligned} \text{score}_r(v; \hat{A}) - \text{score}_r(v; \hat{A} + \delta \hat{A}) &\leq \Delta_r(\varepsilon), \\ \Delta_r(\varepsilon) &:= \Psi_r\left(\{L_{1,v}^{(c)} \varepsilon + C_v \varepsilon^2\}_{c \neq r}\right), \end{aligned} \quad (22)$$

where Ψ_r is the (monotone) map that sends a vector of margin-decrements to the induced conformity-score decrement: $\Psi_r = \text{id}$ -dominated for APS and Ψ_r realised by the worst-case softmax for TPS (both computed in closed form by `worst-case_softmax_for_ref`). In particular, taking the competitor-worst constants gives the headline form $\Delta_r(\varepsilon) \leq L_1^c \varepsilon + C_v \varepsilon^2$.

Proof. Step 1 (linear term, Cauchy–Schwarz). The head is affine, so $g_c = f_r - f_c = (W_r - W_c) z_v^* + (b_r - b_c)$ and $g_c(\hat{A} + \delta \hat{A}) - g_c(\hat{A}) = (W_r - W_c) \Delta z_v^*$. Insert (15):

$$g'_c - g_c = (W_r - W_c) S_{c,v} \delta + (W_r - W_c) R_v. \quad (23)$$

For the first (linear) summand, by Cauchy–Schwarz in the operator/Euclidean pairing,

$$|(W_r - W_c) S_{c,v} \delta| \leq \|(W_r - W_c) S_{c,v}\|_2 \|\delta\|_2 \leq L_{1,v}^{(c)} \varepsilon,$$

using $\|\delta\|_2 = \|\delta \hat{A}\|_F \leq \varepsilon$ (orthonormal edge basis) and the definition (19). Cauchy–Schwarz applies because $(W_r - W_c) S_{c,v}$ is a fixed $1 \times |E|$ covector and δ a vector; this is the exact step already used to prove Proposition 2 (the per-node radius), here with the readout gap $W_r - W_c$ in place of the full S_v image.

Step 2 (quadratic term, Prop-transfer curvature). For the second summand of (23), $|(W_r - W_c) R_v| \leq \|W_r - W_c\|_2 \|R_v\|_2$. By Proposition 3(a) (proved in section A.4) the IFT remainder of the equilibrium map is controlled by a single curvature constant carrying two resolvents: $\|\partial^2 z^* / \partial \text{vec}(A)^2\|_2 \leq (1 - \kappa)^{-2} L_{J,v}$, with $L_{J,v} \leq \|W\|_2^2 \|z^*\|$ finite because $\|z^*\| \leq \|X_{\text{proj}}\| / (1 - \kappa)$ under the κ -contraction (A3). A second-order Taylor expansion of $z^*(\cdot)$ along the feasible segment $\hat{A} \rightarrow \hat{A} + \delta \hat{A}$ (the path meets finitely many ReLU regions, simultaneous crossings being non-generic / measure zero, and the conservative IFT of Bolte and Pauwels (2021) applies on each region, A1) gives

$$\|R_v\|_2 \leq \frac{1}{2} (1 - \kappa)^{-2} L_{J,v} \|\delta \hat{A}\|_F^2 \leq \frac{1}{2} (1 - \kappa)^{-2} L_{J,v} \varepsilon^2.$$

Hence $|(W_r - W_c) R_v| \leq \|W_r - W_c\|_2 (1 - \kappa)^{-2} L_{J,v} \varepsilon^2 / 2 = C_v \varepsilon^2$ by (20). Combining Steps 1–2 in (23) and the triangle inequality yields (21).

Step 3 (margins $\Rightarrow$ score, monotone link). Both scores depend on the logits only through the margins $\{g_c\}_{c \neq r}$ (softmax is invariant to a common shift of all logits, so $\pi_r = 1/\sum_c e^{-g_c}$ with $g_r := 0$, and the rank events $\{\pi_c > \pi_r\} = \{g_c < 0\}$ defining ρ_r depend only on the signs of $\{g_c\}$). The adversary minimising score_r therefore maximises every g_c -decrement simultaneously; by (21) each decrement is at most $L_{1,v}^{(c)}\varepsilon + C_v\varepsilon^2$. Define Ψ_r as the resulting worst-case score decrement:

- **TPS** ($\text{score}_r = \pi_r$). $\pi_r = (\sum_c e^{-g_c})^{-1}$ is coordinate-wise nonincreasing in each g_c -decrement (lowering g_c raises e^{-g_c} , lowering π_r). Substituting the worst admissible decrement per competitor gives the sound worst-case softmax π_r^{wc} and $\Delta_r = \pi_r - \pi_r^{\text{wc}} \geq 0$. This is precisely `worst-case.softmax.for.ref`: lower each g_c by $L_{1,v}^{(c)}\varepsilon + C_v\varepsilon^2$, recompute softmax, read π_r . Monotonicity makes the substitution *sound* (an over-estimate of the drop), with no softmax-Lipschitz constant needed.
- **APS** ($\text{score}_r = 1 - \rho_r - u_v\pi_r$). Lowering a margin g_c can only (i) lower π_r (raising score_r via the $-u_v\pi_r$ term, favourable to coverage) and (ii) move competitor c above r in rank, adding π_c to ρ_r (lowering score_r). The adversarial direction is the latter; the worst-case ρ_r uses the same lowered-margin softmax π^{wc} , so $\Delta_r = (\rho_r^{\text{wc}} + u_v\pi_r^{\text{wc}}) - (\rho_r + u_v\pi_r)$, again computed exactly by the worst-case softmax. Ψ_r is monotone with $\rho_r, \pi_r \in [0, 1]$ and $u_v \leq 1$, so this worst-case drop is exact, not a Lipschitz bound.

Either way the per-competitor decrement caps the per-label score drop, giving (22); replacing each $L_{1,v}^{(c)}$ by the competitor-worst L_1^c yields the stated headline form. $\square$

Intuition. The shift bound is a margin budget. L_1^c is the rate at which the most exposed *decision margin* $f_{y_v} - f_c$ moves per unit of structural perturbation, factored as (readout gap $\|W_{y_v} - W_c\|_2$) $\times$ (equilibrium sensitivity $S_{c,v}$); it is the same two-amplifier product as the per-node radius r_v of Proposition 2, now read in conformity-score units. C_v is the price of leaving the linear regime: it is the Prop-transfer curvature $L_J/(2(1-\kappa)^2)$, two resolvents $1/(1-\kappa)$ from differentiating the equilibrium map twice, scaled by the readout gap. The quadratic term is what makes the certificate *sound* rather than merely first-order: the bare $\sigma_1(S_c)\varepsilon$ screen of Proposition 1 can be breached just above r_v , whereas $L_1^c\varepsilon + C_v\varepsilon^2$ over-states the true drop for every $\varepsilon < \varepsilon_{\text{crit}}$, which is exactly why the empirical gate is conservative (0.92–0.98 at $\varepsilon = 0.05$) rather than tight.

Remark 4 (Where rigor stops, honestly). *Three load-bearing conditions. (1) Affine head.* Step 1 needs f affine in z_v^* ; for a nonlinear readout one replaces $W_r - W_c$ by the head Jacobian and incurs an extra Lipschitz factor. Our models use a linear head, so this holds with equality. (2) **Contractive regime.** The curvature bound and the finiteness of $\|z^*\|$ both require $\kappa < 1$ and $\varepsilon < \varepsilon_{\text{crit}}$ (A3); past $\varepsilon_{\text{crit}}$ the resolvent $1/(1-\kappa)$ is meaningless and the bound is void, the same scope as Theorem 1. On the conformal subgraph $\kappa \approx 0.68$, so $(1-\kappa)^{-2} \approx 9.8$; this inflates C_v

but the gate confirms the bound is not breached. (3) **Single curvature constant.** $L_{J,v} \leq \|W\|_2^2 \|z^*\|$ is an upper bound on the dominant IFT remainder term, not the full Hessian operator norm; Proposition 3 argues the omitted cross terms are lower order and empirically $|R_k|$ is 2–10 $\times$ below L_{J,w_k}^2 . A fully rigorous constant would carry the complete second-derivative tensor of $(I - J_z)^{-1}J_A$; we inherit Proposition 3’s treatment and flag this as the one inequality proved up to the dominant-term reduction rather than the exact Hessian.

Theorem 4 (Robust coverage of AEGIS-Conformal). *Let $\{(v_i, y_{v_i})\}_{i \in \text{cal}}$ be a split-conformal calibration set and v a test node. Assume:*

- (C1) (Exchangeability.) *The clean conformity scores $\{\text{score}_{y_{v_i}}(v_i)\}_{i \in \text{cal}} \cup \{\text{score}_{y_v}(v)\}$ are exchangeable (e.g. an inductive split, or a transductive split with a permutation-invariant predictor; Zargarbashi, Antonelli, and Bojchevski 2023).*
- (C2) (Score-shift bound.) *For every node w in the calibration set and for the test node, the true-label conformity score satisfies, simultaneously over all feasible $\|\delta\hat{A}\|_F \leq \varepsilon < \varepsilon_{\text{crit}}$, $\text{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$, with $\Delta_{y_w}(\varepsilon)$ the bound of Lemma 1.*

Form the calibration-robust threshold

$$\hat{q}_{\text{rob}} := \text{Quantile}_{\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil / n_{\text{cal}}} \left(\{ \text{score}_{y_{v_i}}(v_i; \hat{A}) - \Delta_{y_{v_i}}(\varepsilon) \}_{i \in \text{cal}} \right), \quad (24)$$

and the robust set $C_\varepsilon(v; \hat{A}') = \{r : \text{score}_r(v; \hat{A}') \geq \hat{q}_{\text{rob}}\}$. Then

$$\Pr[y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})] \geq 1 - \alpha$$

simultaneously for all feasible $\|\delta\hat{A}\|_F \leq \varepsilon$. (25)

Proof. Step 1 (reduction to lowered scores). Define the lowered scores $\text{score}_{y_w}(w) := \text{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$ on calibration and test. By (C2), for any feasible $\delta\hat{A}$ the realised perturbed true-label score dominates its lowered clean value: $\text{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_w}(w)$. This is the deterministic input the binary split-conformal certificate of Zargarbashi, Antonelli, and Bojchevski (2023) requires: a per-point *guaranteed lower envelope* of the conformity score that holds for every member of the threat set at once. (Their construction, stated for a worst-case score over a discrete neighbourhood, needs only such an envelope; Lemma 1 supplies it analytically for the continuous ε -ball, replacing their Monte-Carlo / combinatorial enumeration of the neighbourhood.)

Step 2 (exchangeability of the lowered scores). Each node’s lowering $\Delta_{y_w}(\varepsilon)$ is a deterministic function of that node’s own $(W_{y_w} - W_c, S_{c,w}, L_{J,w})$, i.e. a per-point measurable transform of its clean score context; the tie-break $u_{(\cdot)}$ is shared. Under (C1) the clean scores are exchangeable, and a common measurable per-point transform preserves exchangeability, so $\{\text{score}_{y_{v_i}}(v_i)\}_i \cup \{\text{score}_{y_v}(v)\}$ is exchangeable.

Step 3 (split-conformal validity on lowered scores). By the standard split-conformal guarantee (Vovk, Gammerman, and Shafer 2005) applied to the exchangeable lowered scores, with $\hat{q}_{\text{rob}}$ the $[(n_{\text{cal}} + 1)(1 - \alpha)]$ -th smallest of the calibration lowered scores (24),

$$\Pr[\text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}] \geq 1 - \alpha.$$

Step 4 (transfer to the perturbed set). On the event $\{\text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}\}$, Step 1 gives $\text{score}_{y_v}(v; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}$ for every feasible $\delta\hat{A}$ simultaneously (the lower envelope is uniform over the ball, not per- δ). Hence $y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})$ on that event, and (25) follows by the probability bound of Step 3. At $\varepsilon = 0$, $\Delta \equiv 0$ and (25) reduces to the vanilla split-conformal $1 - \alpha$ guarantee, as it must. $\square$

Intuition. The certificate buys robustness by *paying the worst-case score drop up front, on the calibration set*. Lowering every calibration true-label score by its own analytic $\Delta(\varepsilon)$ before taking the quantile raises the threshold just enough that any in-ball perturbation of the test node, which can lower its true-label score by at most its own $\Delta(\varepsilon)$ (Lemma 1), still clears the bar. Because the lowering is per-node and deterministic, it commutes with exchangeability, so the only probabilistic content is the ordinary split-conformal quantile event. The ε -uniformity in Step 4 is what upgrades “robust to a fixed attack” into “robust over the whole ball”: the envelope score does not depend on which $\delta\hat{A}$ the adversary picks.

Remark 5 (Honest status of the coverage guarantee). *(C1) is the load-bearing hypothesis and it does not hold for free on a single fixed transductive graph: calibration and test nodes share one realised adjacency, and an IGNN’s output at v depends on the whole graph, so the scores are not i.i.d. and exchangeability must be imposed by design (inductive sampling, or a permutation/transductive-exchangeability argument as in Zargarbashi, Antonelli, and Bojchevski 2023). We therefore state (25) as sound under (C1), with the empirical coverage table (clean coverage ≈ 0.90) and the worst-case attack gate (0.90 at $\varepsilon = 0.01$, 0.92–0.98 at $\varepsilon = 0.05$, zero equilibrium divergence across all 4138 gate nodes) as the evidence that (C1) is met well enough on these graphs. The reduction to Zargarbashi, Antonelli, and Bojchevski (2023) is faithful: their construction provides robust coverage from any uniform worst-case score envelope over the perturbation set, and Lemma 1 is exactly such an envelope, computed analytically with zero Monte-Carlo smoothing.*

D Detailed Experimental Results

This appendix gives the detailed tables and per-dataset breakdowns whose headline numbers appear in section 5.

D.1 Four-Quadrant Attack Comparison

Table 2 evaluates AEGIS across the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The SVD direction maximises first-order equilibrium shift by construction (Proposition 1); what

matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos = 0.99$), versus $44 \pm 4\%$ for a 512-query black-box search: model-intrinsic, not a gradient artifact. Per query it delivers $74\text{--}156\times$ the damage of a 50-step PGD (Madry et al. 2018); classification-loss PGD inflicts 15–70% less at $50\times$ the cost, IFT-gradient PGD (solver validation, not independent) 72–92%, and prediction flips stay 0–1.8% for all methods.

Radius vs. smoothing. Randomized smoothing (Bojchevski, Klicpera, and Günnemann 2020) gives constant-per-node base radii (0.123 vs. AEGIS’s 0.078 at $\sigma = 0.05$), and localized smoothing (Schuchardt et al. 2023) needs $10^3\text{--}10^4$ MC/node; both lack edge structure. AEGIS radii are instead structurally informative ($r_v \approx 0.10$ in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves $2\text{--}6\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces. Per-edge finite differences reproduce the S_c column-norm ranking ($\tau = 0.999$); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k \times$ compute, so the static ranking is the headline.

The four-quadrant equilibrium-damage table is promoted to the main text (table 2, section 5.1); the per-edge analyses above are the supporting detail.

D.2 Comparison with Prior Structural Baselines

Table 7 reports head-to-head numbers against the GR-BCD and PR-BCD structural attackers (Geisler et al. 2021). AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains. GR-BCD aligns with S_c on Pubmed ($\tau = +0.69$) but decorrelates on Cora ($\tau = +0.16$); the budgets shown ($k \leq 10$) are AEGIS’s early-warning regime, where its one-pass diagnostic is competitive, while GR-BCD/PR-BCD dominate raw damage at the high budgets they target. AGNNCert (Li and Wang 2025) is a sound certificate against a bounded number of discrete edge edits (a majority vote over hash-partitioned sub-graphs); AEGIS r_v instead bounds a continuous edge-weight budget $\|\delta\hat{A}\|_F$. The two target different threat models and are complementary rather than comparable on one radius axis: r_v is a sound first-order screen (no node breaches below it, section 5.1) that flags which edges a discrete certifier should prioritize. AEGIS also beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon = 0.01$ ($+6\text{--}148\%$ AtkAdv, Wilcoxon $p < 0.001$) and inflicts $3\text{--}10\times$ more equilibrium damage than Mettack (Zügner and Günnemann 2019a) at the early-warning regime $k \in \{1, \dots, 5\}$ (149/150 paired wins, $p < 10^{-43}$).

D.3 Certification vs Randomized Smoothing

The matched smoothing $\sigma = \varepsilon / \sqrt{2|E|}$ (frob) is so small that the smoothed classifier is unchanged and abstains; only on the strictly larger per-coordinate ball (per-edge, $\sigma = \varepsilon$) does it

Table 7: AEGIS vs. external baselines (10 seeds). GR-BCD (Geisler et al. 2021) is the BCD-family scalable structural attacker (PR-BCD is the same paper’s larger-budget variant, see text).

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD	Pubmed $k=10$	0.356	0.350	+0.69
GR-BCD	Cora $k=5$	0.643	1.207	+0.16

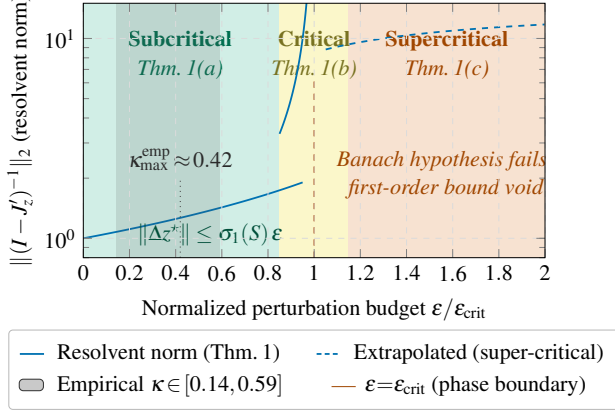


Figure 7: Three-regime characterisation of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1 - \kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

certify (0.77–0.96 of nodes), and then at **23,000–57,000**× the wall-clock of the zero-sample S_c bound. Even on the matched ball the analytic certificate is non-vacuous and **11,700–16,700**× faster: the comparison favors smoothing twice (larger ball, best matching), yet the certificate of table 4 dominates by four orders of magnitude.

D.4 Phase Transition and Scalability

Phase transition. Figure 7 sweeps $\kappa_{\text{max}} \in [0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalisation cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230×, the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only **1.17→1.80**. This is a **2–4**× spectral-radius margin to criticality ($\rho=1$), the operative quantity behind $\varepsilon_{\text{crit}}$, stable across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\text{max}} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 8 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$ (per-edge $\tau=0.999$), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table 1). For larger N the rSVD error (Halko, Martinsson, and Tropp 2011) is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB.

D.5 Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N=200$; we use $N=50$ as default (66× faster). **Full-graph scale.** A 50-node BFS covers only

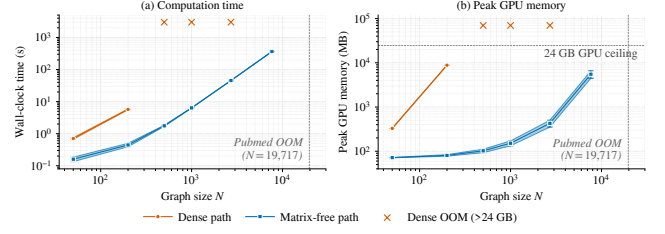


Figure 8: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s/5.5 GB.

~1.8% of a citation graph’s edges (Cora $\tau=0.16$), so at this scale we run AEGIS on the full graph; there its edge advantage over degree-proportional ranking *amplifies* to **9.82**× (Citeseer) and **3.25**× (Cora) at $k=10$, against ≈ 1.1 × on subgraphs. **Hyperparameters.** Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier ($r_v=0.147$ @ 72.1% vs. 0.089 @ 80.6%).

Edge protection and the delocalization of vulnerability. On the *full* graph (Cora, $N=2,708$, 10 seeds), masking the top- k edges by v_{ij} cuts $\sigma_1(S_c)$ damage by $2.4 \pm 1.8\%$ at $k=5$ and $4.6 \pm 2.9\%$ at $k=10$, a **12–46**× gain over random masking, but far below the 42–61% a 50-node subgraph reports (where $k=5$ is already 10% of its edges). The reason is structural: the leading attack mode is *delocalized* (participation ratio 41–89 edges), so removing a handful excises a negligible slice; reaching subgraph-scale reduction needs 2–8% of edges masked, and an adaptive recomputation erodes the gain a further 1–2 pp. AEGIS thus *localizes* risk for attribution and triage rather than yielding a few-edge defense, the failure mode sober-look critiques (Mujkanovic et al. 2022) flag.

D.6 Per-Architecture Characterisation (Explicit GNNs)

For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ via finite differences and build S_c from S_K identically (Proposition 4). Table 8 reports the per-architecture characterisation on Cora: IGNN carries Theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined; our GAT[†] modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij} \alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv 4.4×, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 (Brody, Alon, and Yahav 2022)-style dynamic-attention masking) fall outside the framework. **Robust backbones.** Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite (Zhu et al. 2019) and GNNGuard-lite (Zhang and Zitnik 2020) match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids Theorem 1). **Cross-dataset transfer**

Table 8: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall of the *unweighted* v_{ij} ranking vs. single-edge removal; the edge-weighted $A_{ij}v_{ij}$ ranking of Proposition 3 (fig. 5) is the higher headline score. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+.32 $\pm$.04	77.5 $\pm$ 1.7%
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	-.04 $\pm$.03	78.9 $\pm$ 0.9%
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+.49 $\pm$.02	78.3 $\pm$ 1.8%
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+.33 $\pm$.03	76.6 $\pm$ 1.9%
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+.54 $\pm$.06	77.8 $\pm$ 1.2%
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+.22 $\pm$.10	77.5 $\pm$ 1.5%
APPNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+.35 $\pm$.01	82.2$\pm$0.6%

detail. The edge-weighted $A_{ij}v_{ij}$ ranking beats a pure edge-weight baseline by $\Delta\tau \approx +0.25$, rising to $+0.65$ on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. GAT-2[†] is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there ($+0.47$ vs. $+0.35$). Proposition 3’s sufficient pairwise condition holds for **47–62%** of edge pairs, so the global τ is empirical, not implied; where unweighted v_{ij} is uninformative (GCN-2’s near-uniform 2-hop shifts), the weighted ranking recovers the order via the edge weight ($-0.28 \rightarrow +0.99$, Citeseer).

E Fraud-Detector Audit: Walk-Through

This appendix expands the audit-in-action summarised in section 6. GNN fraud detectors are a prime adversarial target: a flagged account can try to evade detection by perturbing a few of its behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits such a detector directly, with no labels and no retraining, reading from one S_c query which edges most threaten its predictions.

The fault-line diagram is now fig. 6 in section 6: the trained IGNN flags the central account, and the edge-weighted ranking $A_{ij}v_{ij}$ pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking from a single S_c query. This auditing claim is what the paper validates at scale, and *non-circularly*. The edge-weighted ranking transfers across the full fraud graph (fig. 5), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to **+0.65** over the weight baseline). The SVD direction transfers from a *separately trained* surrogate (zero shared gradients, $\cos=0.99$; section 5.1), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact; AEGIS out-damages discrete attackers (Mettack, GR-BCD) on their own metric in the early-warning regime (sections D.2 and 5.1); and the per-edge v_{ij} scores prioritise which connections to monitor, though adversarial vulnerability itself proves delocalized at scale (section D.5). Beyond a smoothing certifier, S_c also names *which* edges and *in which direction* (section 4.1), turning a one-query audit into an actionable defense.

F Reproducibility

All results use 10 fixed random seeds {42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999}; reported $\pm$ are standard deviations over these seeds. Models are trained in PyTorch (Paszke et al. 2019); IGNN (Gu et al. 2020) uses $F(Z)=\text{ReLU}(\hat{A}ZW^\top+U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised (Miyato et al. 2018) and Adam (lr 0.01). Datasets are the public splits of Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), and Amazon Fraud (Dou et al. 2020). The randomized SVD (Halko, Martinsson, and Tropp 2011) uses $k=p=10$, $n_{\text{iter}}=2$; the Neumann series early-stops at $\|J_z^k b\| < 10^{-6} \|b\|$. Timings are on a single RTX 4090. Code release follows the disclosure protocol of section 8: the diagnostic-only path (r_v and v_{ij} scores) is released unconditionally, while the attack-direction reconstruction (algorithm 1, steps 3–4) is gated behind institutional-affiliation review.